Literature-Already-Answered Packet

Vector-valued dimension-free Hardy–Littlewood maximal bounds on the Heisenberg group

Status: literature already answered, not a new automatic-discovery result

Source question

The source paper is arXiv:1703.08327, Deleaval–Kriegler, *Dimension free bounds for the vector-valued Hardy–Littlewood maximal operator*. It proves dimension-free Fefferman–Stein inequalities on Euclidean spaces and for Grushin maximal operators. In the final paragraph, the authors point to the centered Hardy–Littlewood maximal operator on the Heisenberg group and say that they do not know whether this operator admits dimension-free ℓ^q-valued or UMD Banach lattice-valued estimates.

Later answer

The later paper is arXiv:2503.15291, Ganguly–Ghosh, *Dimension free estimates for the vector-valued Hardy–Littlewood maximal function on the Heisenberg group*. The supporting TeX source is included at

`source_tex/supporting_2503.15291.tex`.

Ganguly–Ghosh explicitly cite the Heisenberg-group analogue left open by Deleaval–Kriegler as the motivating open problem. They work on $\mathbb{H}^n$ and define the centered Hardy–Littlewood maximal operator using Koranyi balls.

Claim. *For every $1 < p, q < \infty$, the centered Hardy–Littlewood maximal operator on $\mathbb{H}^n$ satisfies the Fefferman–Stein inequality*

$$\left\| \left(\sum_j |\mathcal{M} f_j|^q \right)^{1/q} \right\|_{L^p(\mathbb{H}^n)} \leq C(p,q) \left\| \left(\sum_j |f_j|^q \right)^{1/q} \right\|_{L^p(\mathbb{H}^n)},$$

where $C(p,q)$ is independent of n.

This is Theorem 1.1 of arXiv:2503.15291.

Claim. *If Z is a UMD Banach lattice and $1 < p < \infty$, then the corresponding UMD-lattice-valued centered Hardy–Littlewood maximal operator on $\mathbb{H}^n$ is bounded on $L^p(\mathbb{H}^n; Z)$ with a constant $C(p,Z)$ independent of n.*

This is Theorem 1.2 of arXiv:2503.15291.

Metric variant

The source question cites earlier Heisenberg maximal-function papers rather than pinning the question to only one Heisenberg ball model. Ganguly–Ghosh's main results are stated for Koranyi balls. Their concluding remarks also state the same dimension-free ℓ^q and UMD-lattice-valued estimates for Hardy–Littlewood maximal averages over Carnot–Caratheodory balls.

Classification

This packet records a later-literature resolution of the source paper's explicit "we do not know" signal. It should not be counted as a new proof by this run.